ACCT102 Management Accounting

Week 4

Job-order costing

Learning Objective

Distinguish between job-order costing and process costing and identify companies that would use each costing method.

Types of Product Costing Systems

Job-order costing

- Many different products are produced each period.
- Products are manufactured to order.
- The unique nature of each order requires tracing or allocating costs to each job, and maintaining cost records for each job.

Examples:

- *Boeing (aircraft manufacturing)*
- *Bechtel International (large scale construction)*
- *Walt Disney Studios (movie production)*

Process costing

- A company produces many units of a single product.
- One unit of product is indistinguishable from other units of product.
- The identical nature of each unit of product enables assigning the same average cost per unit.

Examples:

- *Asia Pulp & Paper (paper manufacturing)*
- *Nissin (instant noodle producer)*
- *Coca-Cola (beverages)*

Job-Order vs Process Costing

	Job-Order	Process
Number of jobs worked	Many	Single Product
Cost accumulated by	Individual Job	Department or Process
Average cost computed by	Job	Department or Process

Quick Check 4A.1

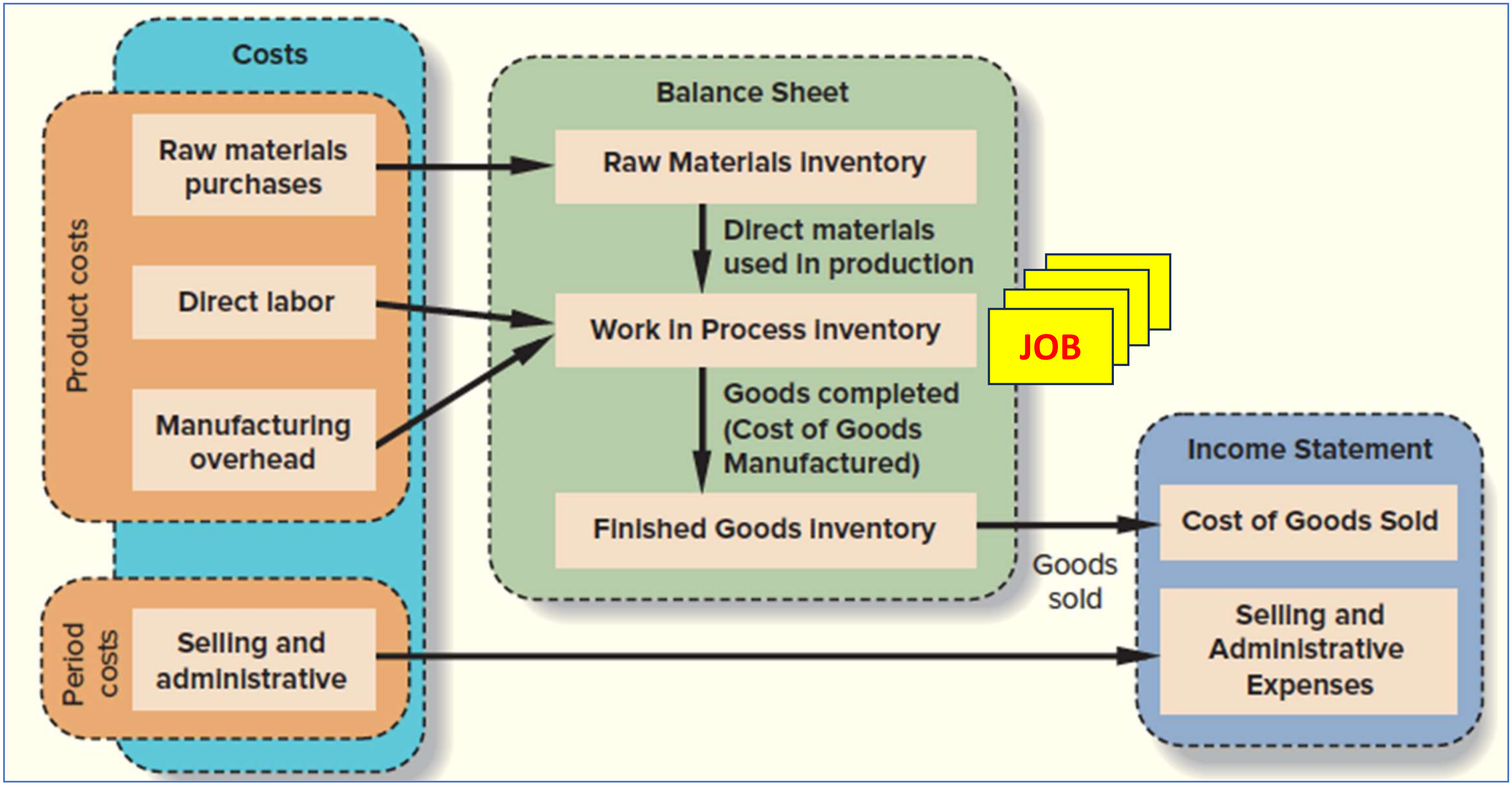
Which of the following company would be likely to use job-order costing rather than process costing?

a. Heinz for ketchup.

b. Caterer for a wedding reception.

Job-Order Costing – Cost Flows

1. A **sales order** is the basis of issuing a **production order**.
2. A **production order** initiates work on a job.



The Job Cost Sheet

PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-XX

Date Completed

Department B3

Units Completed

Item Wooden table

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount

Cost Summary	Units Shipped		
	Date	Number	Balance
Direct Materials			
Direct Labor			
Manufacturing Overhead			
Total Cost			
Unit Product Cost			

Job Cost Sheet – Direct Materials Cost

PearCo

Job Number A - 143

Department B3

Item Wooden table

Direct Materials		Direct	
Req. No.	Amount	Ticket	Ho
X7-6890	$ 116		

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor				
Manufacturing Overhead				
Total Cost				
Unit Product Cost				

PearCo Materials Requisition Form

Requisition No. X7 - 6890 Date 3-4-XX
Job No. A - 143
Department B3

Description	Quantity	Unit Cost	Total Cost
2 x 4, 12 feet	12	$ 3.00	$ 36.00
1 x 6, 12 feet	20	4.00	80.00
			$ 116.00

Authorized Signature *Will E. Delite*

Job Cost Sheet – Direct Labor Costs

PearCo Employee Time Ticket

Time Ticket No. 36 Date 3-5-XX

Employee I. M. Skilled Station 42

Starting Time	Ending Time	Hours Completed	Hourly Rate	Amount	Job No.
0800	1600	8.00	$ 15.00	$ 120.00	A-143
Totals		8.00	$ 15.00	$ 120.00	A-143

Supervisor *C. M. Workman*

t Sheet

Initiated 3-4-XX

Completed

s Completed

Req. No.	Amount	Ticket	Hours	Amount	Manufacturing Overhead Hours	Manufacturing Overhead Rate	Manufacturing Overhead Amount
X7-6890	$ 116	36	8	$ 120			

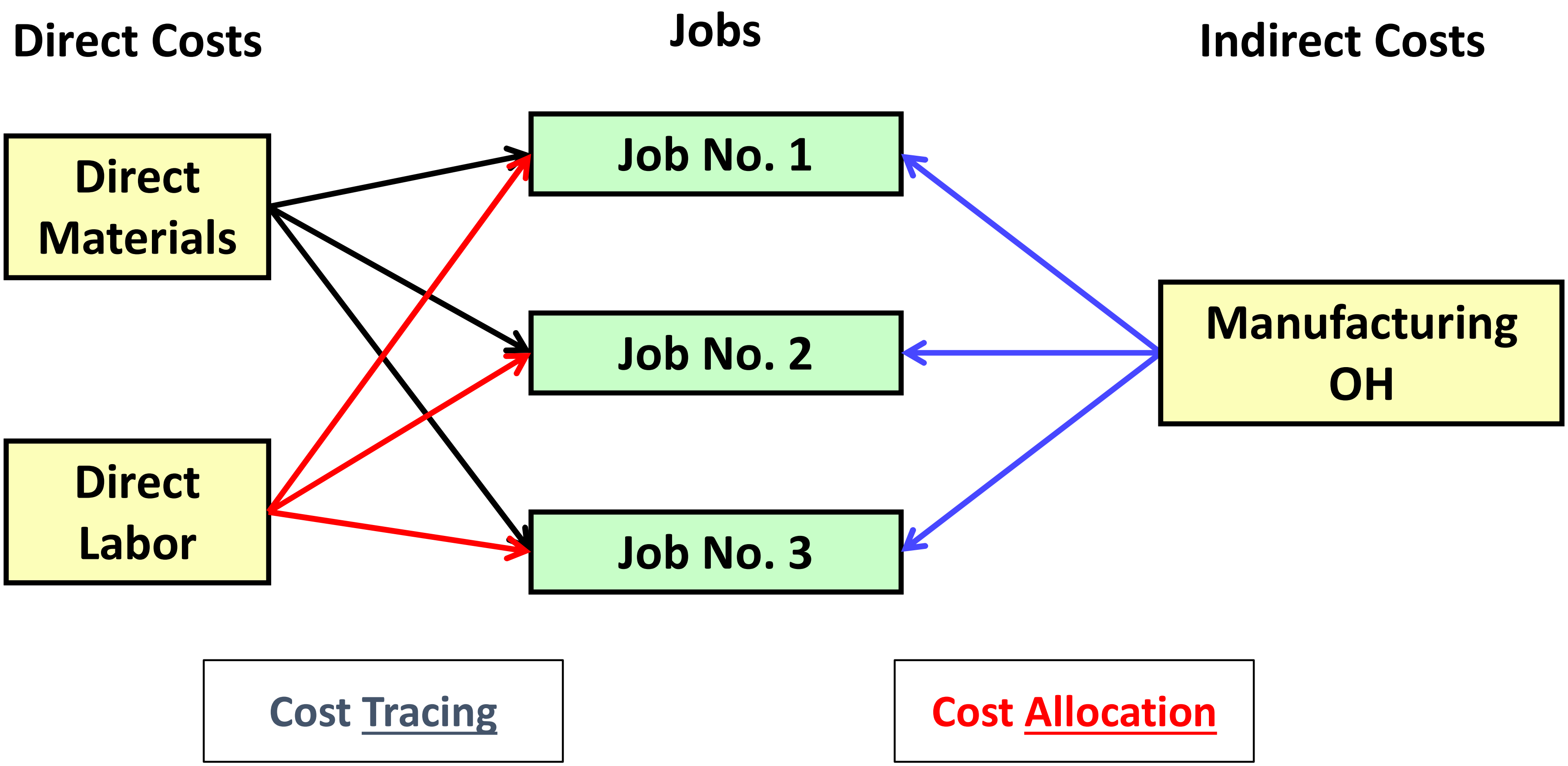
Cost Summary	
Direct Materials	$ 116
Direct Labor	$ 120
Manufacturing Overhead	
Total Cost	
Unit Product Cost	

Units Shipped		
Date	Number	Balance

Allocating Manufacturing Overheads

Charge direct material and direct labor costs to each job as work is performed.

Manufacturing Overhead, including indirect materials and indirect labor, are allocated to all jobs rather than directly traced to each job.



Actual Costing vs Normal Costing

Actual costing – A costing system in which **actual** overhead costs are allocated to products

- **Job-order:** Actual overhead rate multiplied by the actual amount of the allocation base incurred by the job.

Normal costing – A costing system in which overhead costs are applied to a job using a **predetermined overhead rate**

- Job-order: Predetermined overhead rate multiplied by the actual amount of the allocation base incurred by the job.

Why must the overhead rate be predetermined?

- Since Actual Overhead for the period is not known until the end of the period, using a predetermined rate makes it possible to estimate total product/job costs sooner so that correct pricing decisions can be made.

Start of month		End of month
Pricing decision made		Actual overhead cost known

- Actual Overhead costs may have seasonal fluctuations throughout the year and these seasonal effects should be ignored.

Normal Costing – Applying Overheads

Normal costing – A costing system in which overhead costs are applied to a job by multiplying a **predetermined overhead rate** by the **actual amount of the allocation base incurred by the job**.

1. Determine the predetermined overhead rate

A predetermined overhead rate is calculated requiring us to use **estimated/ budgeted** figures for both the cost and business activity.

$$\textbf{Predetermined}\text{ Overhead Rate} = \frac{\textbf{Budget/Estimated}\text{ OH for the period}}{\textbf{Budget/Estimated}\text{ \# of }\textbf{allocation base}\text{ for the period}}$$

Allocation base may be # of units, machine hours, labour hours etc.
Ideally, the allocation base is a cost driver that causes overhead.

Normal Costing – Applying Overheads

2. Apply POHR to actual activity

Based on estimates, and determined before the period begins.

Applied overhead = **POHR** x **Actual activity**

Actual amount of allocation is based upon the actual level of activity ~ per customer, per job, per month

Quick Check 4A.2

Under Normal Costing, predetermined overhead rates are based on actual cost and activity data.

A) True

B) False

Job Cost Sheet – Manufacturing Overheads (1)

- PearCo uses normal costing and a firm-wide predetermined overhead rate.
- Overhead cost mainly driven by direct labor hours.
- Budgeted total manufacturing overheads = $40,000
- Budgeted total direct labor hours = 10,000 DLH

$$\textbf{POHR} = \frac{\text{Estimated total manufacturing overheads of PearCo}}{\text{Estimated total direct labor hours of PearCo}}$$

$$\frac{\$40{,}000}{10{,}000} = \textbf{\$4 per DLH}$$

POHR = $4 per DLH

Job Cost Sheet – Manufacturing Overheads (2)

PearCo Job Cost Sheet

Job Number A - 143	Date Initiated 3-4-XX
	Date Completed 3-5-XX
Department B3	Units Completed 2
Item Wooden table	

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead	$ 32			
Total Cost				
Unit Product Cost				

Calculating Total Job Cost and Unit Product Cost

PearCo Job Cost Sheet

Job Number A - 143

Department B3

Item Wooden table

Date Initiated 3-4-XX

Date Completed 3-5-XX

Units Completed 2

Does not always mean 1 item.

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead	$ 32			
Total Cost	$ 268			
Unit Product Cost	$ 134			

$268 / 2

Interpreting the Unit Product Cost

- Unit Product Cost ≠ Incremental Cost per Product
- The average unit cost should not be interpreted as the costs that would actually be incurred if an additional unit was produced.
- Fixed overhead would not change if another unit was produced, so the incremental cost of another unit is something less than $134.

Quick Check 4A.3

Job WR53 at NW Fab, Inc. incurred $200 of direct materials and $20 of indirect materials. 10 direct labor hours at $15 per hour was incurred and $380 of manufacturing overheads were allocated. Under normal costing, what would be recorded as the cost of job WR53?

a. $350.

b. $220.

c. $750.

d. $730.

Job-Order Costing in Service Industries

- Besides manufacturing applications, job-order costing is also used in service industries. Job-order costing is used in many different types of service companies. For example, law firms, accounting firms, and medical treatment.
- Job Costing in Professional Firms: https://mediacast.smu.edu.sg/media/0_o8jezkwa

End of topic